



# Addressing the Contemporary Challenges of Business Process Compliance

## The Case for Process Mining in the Banking Industry

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**Abstract** The aim of business process compliance (BPC) is to ensure that business processes are executed in accordance with a prescribed set of rules. In practice, the evidence would suggest that achieving this goal is challenging. Penalties and subsequent remediation costs in the Australian banking industry amounted to over A\$10bn between 2017 and 2022. The research community has identified many challenges, but the industry perspective is typically missing from that research. The study takes advantage of recent events in the Australian banking industry that saw detailed regulatory reports made public, highlighting the challenges the industry faces trying to maintain compliance. The study supplements these reports with BPC-related insights from practitioners and consultancy groups to develop an industry perspective on the challenges. Consolidating the two perspectives presents a comprehensive view of BPC's challenges and the differing emphasis that each stakeholder group places on the challenges. Process mining may provide a pathway to reconciling these different perspectives. Both the BPC literature and industry have promoted process mining's potential to address the challenges. To explore this proposition, the study details the features of representative commercial process mining software and then maps these features to the challenges. The resultant conceptual map is used to

analyze known and novel BPC challenges. It outlines the limitations that must be addressed, from both a research and industry perspective, to maintain and improve compliance.

**Keywords** Business process compliance · Process mining · Conformance checking · Banking · Australia

### 1 Introduction

Business process compliance (BPC) aims to ensure that business processes are executed in line with a prescribed set of rules or norms (Governatori and Sadiq 2009). The evidence suggests that this is challenging in practice. Between 2017 and 2022, Australian regulators issued two of the four major domestic banks with penalties exceeding A\$2bn for breaching the AML/CTF Act<sup>1</sup> (APRA 2018; Westpac 2020), a Royal Commission was held into misconduct in the industry (Hayne 2019), and the banks spent more than A\$8bn on remediation (Adams et al. 2022).

Academic interest in the field of BPC traces its roots to corporate scandals at organizations such as Enron, HIH, AIG, and Société Générale and the ensuing legislative changes (e.g., Dodd-Frank, Sarbanes-Oxley) at the turn of the millennium (Rosemann and zur Muehlen 2005). The research community has recognized that there are many challenges associated with BPC (Hashmi et al. 2018). For example: (i) fully automating BPC may be beyond reach (Barnawi et al. 2016); (ii) identifying and expressing natural language compliance requirements is complex (Governatori and Sadiq 2009); (iii) “compliance by design” (Governatori and Sadiq 2009) is a goal, yet not all

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<sup>1</sup> Anti-Money Laundering and Counter-Terrorism Financing Act.

compliance requirements can be evaluated at design-time (Maggi et al. 2011); and (iv) some of the proposed solutions demand a level of technical expertise unlikely to be found in a commercial setting (Zasada et al. 2023).

These challenges reflect the research perspective, which typically does not necessarily capture the industry perspective, and particularly that of the industry's regulators. Detailed information on compliance breaches is normally confidential, however, the events in the Australian banking industry described in the opening paragraph have resulted in four regulatory reports being published. These reports provide a rare insight into the BPC challenges faced by industry, and using Australian banking as the context, enabled us to study the challenges from the industry perspective. The reports highlight the complexity of the business process environment, and the multitude of organizational challenges that inhibit the banks' ability to maintain compliant business processes.

BPC is acknowledged to be a multi-faceted problem (Barnawi et al. 2016). While both researchers and industry recognize that BPC is challenging, a survey identifying a comprehensive list of the challenges, the differences in emphasis each audience places on them, and any attempt at reconciling these differences, is lacking. Consequently, despite the extent of the research and a technology sector targeting regulatory monitoring, reporting and compliance problems (RegTech), challenges remain, and a comprehensive solution is still to be developed (Sackmann et al. 2018; Deloitte 2022).

The issue can be illustrated by considering conformance checking as a set of process mining functions that compare process instances with a given process model (Dunzer et al. 2019). Conformance checking is covered in the BPC literature as a compliance auditing approach that can also be applied at run time (Hashmi et al. 2018), but not at design time (Carmona et al. 2018). While Hashmi et al. (2018) note the terms "compliance" and "conformance" are used "quasi-synonymously", there are important differences. Conformance checking checks that an event log conforms to a given process model (Dunzer et al. 2019) but does not verify whether the process model adequately reflects the requisite compliance requirements. Hence, a log may conform to a prescribed model but may not comply. The difference is shown in Fig. 1. Furthermore, in practice, not all business processes have fully pre-specified models, and it is not always possible to model the required behavior (Reichert and Weber 2012). From a research perspective, conformance checking does not constitute a comprehensive BPC solution.

However, one advantage of conformance checking is that it has already been commercialized as one of the key process mining capabilities sought out by industry (Gartner 2022). This is important because, not only are these

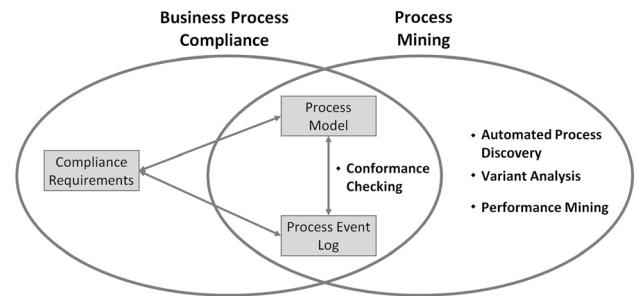


Fig. 1 BPC overlap with process mining

solutions designed for commercial use, but they are already using other process mining techniques to address some of the limitations of conformance checking. For example, segregation of duty-type requirements can be checked by querying a discovered process model using event attributes. From an industry perspective, only a partial commercial solution to BPC is available, which is only effective in the right circumstances, or requires workarounds to check compliance.

This leads to our primary research question: "How can commercial process mining solutions be enhanced to provide a pathway that reconciles the different BPC challenges experienced both by industry and research?" Our response is derived from answering two subsidiary research questions:

- RQ1 What are BPC's challenges from an industry and research perspective, and how do these perspectives differ?
- RQ2 What are the limitations of commercial process mining solutions that must be addressed to provide a pathway to reconcile the different perspectives?

To address RQ1 we applied both a research and an industry lens. For the research lens, we reviewed the BPC literature to identify the challenges recognized by the research community. The industry lens comprised the views of the regulators, consulting groups and industry practitioners. We analyzed the published regulatory reports and insights from consultancy groups active in the Australian banking domain. We supplemented this with our prior work to establish the industry practitioners' views. This constitutes the first part of our contribution – a comprehensive list of contemporary BPC challenges (BPCCs), highlighting the respective areas of focus for industry and the research community.

To address RQ2, we leveraged existing market analysis to extract the key features of commercially available process mining software and assessed the capabilities of five major vendor solutions relative to these features. We then mapped the solution features to the BPCCs to determine if a feature or combination of features could address each

challenge and analyzed the results to identify the limitations. These results form the second part of our contribution. The final part of our contribution is captured in the discussion on the limitations of the solutions to address these challenges, how changing the paradigm can potentially overcome these limitations, and the implications for both research and practice.

While this study is set in an Australian banking context due to the availability of the regulatory reports, given the extent of the penalties issued worldwide, it is unlikely that these findings should be confined to the Australian banking context.

The remainder of this paper is organized as follows. In Sect. 2, we discuss the framework and context of the study, and in Sect. 3 we introduce our methodology. In Sect. 4, we present our results, and we discuss our findings in Sect. 5. In Sect. 6, we present our conclusions and directions for future research.

## 2 Framework and Context

To address our research question, an understanding of both the BPC literature, process mining literature and the Australian banking context is required. Here, we provide a summary of all three.

### 2.1 Business Process Compliance

Governatori and Sadiq (2009) describe BPC as a relationship between what a business must do (the compliance specification) and how a business performs its activities (the process specification). These two sets of specifications have different objectives, stakeholders, and lifecycles – a compliance management lifecycle (Ramezani et al. 2011) and a business process management (BPM) lifecycle (Dumas et al. 2018).

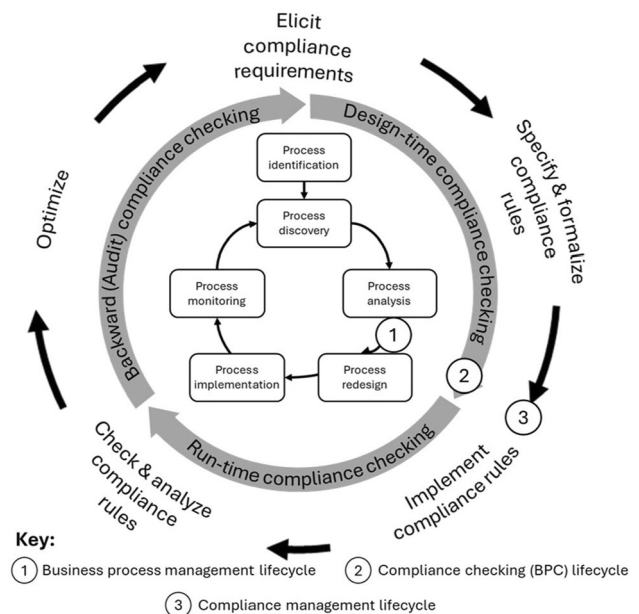
The two lifecycles may interact at different phases, for example, a change in compliance requirements may trigger the process identification phase to determine which business processes are impacted by the change, however, the primary point of convergence is when a representation of the compliance norms is verified against a formal specification of the process (Governatori and Sadiq 2009).

El Kharbili et al. (2008) outline two verification approaches: forward compliance checking and backward compliance checking (audit). They further distinguish between forward compliance checking techniques at design time and those at execution time. The forward compliance checking approaches are pro-active and aligned to the concept of “compliance by design” (Sadiq et al. 2007), whereas backward compliance checking techniques are reactive, detecting violations after they have occurred.

Subsequent BPC studies refer to design time, run time and audit as the BPC lifecycle phases (Ly et al. 2012; Barnawi et al. 2016), albeit the terms may differ, e.g., audit checking may be referred to as post execution or a posteriori checking. The relationship between the three lifecycles is depicted in Fig. 2.

Design-time compliance checking techniques focus on guiding the user during the process model development stage or verifying a designed, but not yet deployed process model (El Kharbili et al. 2008). Debate has centered on approaches and languages that are expressive enough to handle the range and complexity of compliance requirements (Hashmi and Governatori 2018), but are seen to be technically complex, and more user-friendly approaches, applying graphical notations, that potentially lack some of the expressiveness (Elgammal et al. 2010; Reichert and Weber 2012; Zasada et al. 2023). Model checking can also be computationally complex, and checking some models may not be possible (Hashmi et al. 2018).

Run-time methods verify compliance during the process execution phase, and typically address aspects of BPC that cannot be verified and validated at design-time, e.g., segregation of duties and deadlines for completion (Maggi et al. 2011). Proposed solutions fall into two broad categories: reactive, where compliance verifies progress-to-date (Maggi et al. 2011); and proactive monitoring, where progress-to-date knowledge is used to predict compliance outcomes (Rinderle-Ma et al. 2023). Online conformance checking is an example of a reactive solution, discussed in more detail in Sect. 2.2. While there are a range of



**Fig. 2** The BPC lifecycle and its relationship to the compliance management and BPM lifecycles

techniques to capture the run-time process data, event streams are becoming the dominant approach (Maggi et al. 2011).

Auditing is a post-execution approach that has traditionally been both manual and based on sampling (Imran et al. 2023). The focus is now moving to continuous auditing (van der Aalst et al. 2010). Some of the auditing approaches covered in the BPC literature are based on process mining techniques (van der Aalst et al. 2010; Ramezani et al. 2012; Doganata and Curbera 2009). More recent studies have proposed blockchain technology as an auditing approach (Meironke et al. 2019; Viriyasitavat et al. 2022). As a detective control, auditing does not prevent compliance breaches but can be useful to inform process enhancements and assess the impact of changed requirements (Reichert and Weber 2012).

El Kharbili et al. (2008) conclude that an integrated approach to compliance checking is required, covering all phases of the BPC lifecycle. The integrated approach should be intuitive for business analysts and should check for more than just control-flow compliance. In later work, El Kharbili (2012) included all process perspectives: (i) control flow, (ii) time, (iii) resource, and iv) data. Most contributions in the literature are skewed towards forward compliance checking (68.9% of studies analyzed) and formalizing compliance requirements (59.5% of studies analyzed), and there does not appear to be a comprehensive, “bundled” approach (Sackmann et al. 2018).

A number of frameworks have been developed to outline the elements of a BPC solution. Some are specific to a BPC phase, e.g., Ly et al. (2013); Hashmi and Governatori (2018), whereas the Regulatory Compliance Management (RCM) framework in El Kharbili (2012) covers all BPC phases. The elements of the RCM framework are grouped into two dimensions: business process management and compliance, with the compliance dimension further broken down into: (i) modeling, (ii) compliance checking, (iii) compliance analysis (reporting and explanation), and (iv) compliance enactment (violation management). This study’s evaluation of solutions also included a *Business Support* criteria category and a range of functional and non-functional requirements as criteria.

Each element within these frameworks represents a problem that a comprehensive BPC solution should address. While much progress has been made in BPC research, given the scale and diversity of compliance requirements and the fact that these requirements change frequently, BPC is a complex problem area (Sadiq et al. 2007), and there are outstanding challenges to apply the techniques in real-world scenarios (Hashmi et al. 2018; Becker et al. 2012). The goal of automating BPC may be out of reach, with the focus now moving to facilitation rather than full automation (Barnawi et al. 2016). The

challenges are explored in more detail in subsequent sections.

## 2.2 Process Mining

The goal of process mining is to design methods and techniques that can automatically analyze process execution data, i.e., process event logs, to facilitate business process management activities by extracting actionable process knowledge (Dumas et al. 2018). For the past 25 years, the research community has predominantly focused on techniques for addressing (i) automated process model discovery; (ii) process conformance (and compliance) checking; (iii) variant analysis; and (iv) process monitoring.

Automated process discovery techniques aim to analyze process event logs and generate a representation of the process behavior as either a Petri net or a BPMN model (Dumas et al. 2018). The challenge is to design a technique that can systematically discover an accurate and simple process model, which is proving to be elusive (Augusto et al. 2019).

Conformance checking aims to detect differences between the prescriptive behavior captured in a process model and the observed behavior of that process, as recorded in the process’ event log (van der Aalst 2016; Carmona et al. 2018). Carmona et al. (2018) suggest that normative behavior in the finance sector can be modeled and assessed by means of conformance checking. There are known limitations when conformance checking is applied in practice: i) while there are examples of multi-perspective conformance checking (Mehr 2024), most approaches focus on the analysis of the control flow perspective of the process behavior or rely on the existence of a normative process model as input, e.g. in BPMN or Petri nets format, or a declarative process specification (Leemans et al. 2018; Augusto et al. 2020); ii) they can be computationally expensive, with slow execution times, due to the theoretical complexity of the underlying algorithms and the persistent growth in the size of event logs (Carmona et al. 2022).

Conformance checking predominantly operates on historical process execution data, however, it is worth mentioning that some research efforts have focused on the design of online conformance checking (van Zelst et al. 2019). Online conformance checking is a backward-looking, run-time approach. It compares, in real time, completed process execution data from a partial trace against prescriptive process behavior (Carmona et al. 2018).

Process monitoring techniques aim to analyze real-time process execution data, i.e., while the process unfolds, but unlike online conformance checking, they are forward looking (Verenich et al. 2019). There are three types: (i) explorative monitoring aims to compare run-time data



with cases completed earlier; (ii) predictive monitoring aims to provide a prediction regarding the future of the process execution, e.g., what will be the process outcome, how long will the process take to complete?; and (iii) prescriptive monitoring aims to recommend a course of action during process execution to achieve a certain outcome, e.g., if this task were executed, the process execution would reach a positive outcome (van der Aalst 2016).

Process variant analysis techniques compare process behavior by comparing two (or more) event logs from the same process to detect differences in their behavior. The most recent literature review on process variant analysis (Taymouri et al. 2021) identified more than thirty techniques. Most techniques can only detect behavioral differences at the control-flow and process performance level (Taymouri et al. 2021), with the only exception being the solution proposed by (Nguyen et al. 2018), which can detect statistically significant differences also at the process resources and data level.

### 2.3 Australian Banking Context

Between 2017 and 2021, multiple compliance breaches by the major Australian banks have been made public, highlighting the range of challenges they face in trying to maintain process compliance. In response the Australian government instigated a Royal Commission into Misconduct in the Banking, Superannuation and Financial Services Industry (Hayne 2019). At the same time AUSTRAC – a government agency – conducted a series of investigations into the mishandling of financial crime requirements. As a consequence: (i) regulators issued penalties exceeding A\$2 billion (Adams et al. 2022); (ii) banking executive accountability was tightened (Adams et al. 2022); (iii) the banks incurred more than A\$8 billion in remediation costs (Adams et al. 2022); and (iv) three CEOs and two Chairmen of the four major banks resigned (Adams et al. 2022). The findings of the Royal Commission and other regulators include: (i) the extent of legislation banks are subject to (Hayne 2019); (ii) the difficulties banks have both understanding and interpreting the legislation (Hayne 2019); (iii) blurred lines of accountability and bureaucratic decision-making (APRA 2018); (iv) the extent of processing and administrative errors (Hayne 2019); (v) the age and complexity of product systems (APRA 2018; Westpac 2020); (vi) a reactive approach to operational risk management and a reliance on manual, detective controls that do not operate end-to-end (APRA 2018; Westpac 2020); (vii) the inability to detect and address systemic issues in a timely manner (ASIC 2018); and (viii) data is captured in a fragmented way, making it difficult for staff to identify and fix potential issues within the organization (ASIC 2018).

Access to this level of detailed information, which would typically be considered confidential, enabled this study.

## 3 Methodology

To answer our research questions, we proceeded in three phases: (i) the identification and analysis of BPCCs from both a research and industry perspective; (ii) the identification of commercial process mining solutions, their common features and vendor specific capabilities; and (iii) mapping the process mining solutions' features to the identified BPCCs. Below, we elaborate on each of these phases.

### 3.1 Phase 1: Identify BPCCs

The objective of this phase was to identify BPC challenges from both a research and industry perspective, to consolidate the research and industry challenges into a single, comprehensive list of BPCCs and to determine the relative importance of each BPCC by perspective. The phase comprised three stages. Stage one identified the challenges from the research perspective. Stage two identified the challenges from the industry perspective, where we sought the views of those in the industry (practitioners), those regulating the industry, and those advising the industry, i.e., the consultancy groups working with practitioners in the Australian banking industry. Stage three analyzed the two sets of challenges, consolidating them into a single list of BPCCs and calculating the relative importance of each BPCC by perspective.

#### 3.1.1 Stage 1 – Research Perspective

The first stage in this phase was to understand the problem space from a research perspective. BPC is a well-researched area, and there have been multiple studies that have developed frameworks and evaluation criteria to compare alternative solutions, identified specific BPC problems, or proposed functionality and features required of a BPC solution, in essence, the problems a BPC solution should address. Hence, our first step was to develop a comprehensive list of the problems identified in the literature and what made them challenging to solve. Therefore, our search strategy targeted BPC studies that included an evaluation of BPC solutions, such as surveys or literature reviews that summarized the state-of-the-art. For those papers that met the inclusion criteria, our quality criteria sought to identify studies containing tables and/or lists of criteria, features, and challenges. Our search strategy and its results are detailed in Table 1. The selected research studies are shown in Table 2.

**Table 1** Search strategy for identifying BPCCs

| Step | Item                                                                                                                                                                                                                 |                       |           |            |              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------|------------|--------------|
| 1    | Select search sources: Google Scholar (GS), Scopus (Sc), Web of Science (WoS)                                                                                                                                        |                       |           |            |              |
| 2    | Select search terms                                                                                                                                                                                                  |                       |           |            |              |
|      | 2a) Specify publication date range: 2000-2024 (8-Sep-24)                                                                                                                                                             |                       |           |            |              |
|      | 2b) Identify "Keywords" search terms: "business process compliance"                                                                                                                                                  |                       |           |            |              |
|      | 2c) Identify "inTitle" search terms: "challenges" OR "comparative" OR "comparison" OR "evaluation" OR "literature review" OR "state-of-the-art" OR "survey" OR "systematic"                                          | <b>Number of hits</b> |           |            |              |
| 3    | Search each source with the keywords and one of the inTitle search terms:                                                                                                                                            | <b>GS</b>             | <b>Sc</b> | <b>WoS</b> | <b>Total</b> |
|      | 3a) Keywords: "business process compliance" AND inTitle: "challenges"                                                                                                                                                | 38                    | 6         | 4          | 48           |
|      | 3b) Keywords: "business process compliance" AND inTitle: "comparative"                                                                                                                                               | 10                    | 1         | 2          | 13           |
|      | 3c) Keywords: "business process compliance" AND inTitle: "comparison"                                                                                                                                                | 5                     | 1         | 0          | 6            |
|      | 3d) Keywords: "business process compliance" AND inTitle: "evaluation"                                                                                                                                                | 29                    | 5         | 3          | 37           |
|      | 3e) Keywords: "business process compliance" AND inTitle: "literature review"                                                                                                                                         | 35                    | 4         | 3          | 42           |
|      | 3f) Keywords: "business process compliance" AND inTitle: "state-of-the-art"                                                                                                                                          | 15                    | 5         | 2          | 22           |
|      | 3g) Keywords: "business process compliance" AND inTitle: "survey"                                                                                                                                                    | 18                    | 3         | 1          | 16           |
|      | 3h) Keywords: "business process compliance" AND inTitle: "systematic"                                                                                                                                                | 54                    | 7         | 4          | 65           |
|      | <b>Total number of hits.</b>                                                                                                                                                                                         | <b>204</b>            | <b>32</b> | <b>19</b>  | <b>255</b>   |
| 4    | Remove 89 duplicated studies                                                                                                                                                                                         |                       |           |            | 166          |
| 5    | Remove authors' own study                                                                                                                                                                                            |                       |           |            | 165          |
| 6    | Remove 63 studies with less than five citations                                                                                                                                                                      |                       |           |            | 102          |
| 7    | Review the study title and abstract, removing 36 studies and retaining only those that meet the inclusion criteria: studies that fall within the BPC domain, are more than 4 pages in length, are written in English |                       |           |            | 66           |
| 8    | Review the manuscript, removing 36 studies and retaining only those that meet the quality criteria: the body of the study contains a list of BPC features, element, criteria, problems, or challenges                |                       |           |            | 30           |
| 9    | Conduct a backward search and identify two studies to add in that meet the inclusion and quality criteria.                                                                                                           |                       |           |            | 32           |
| 10   | <b>Total studies selected</b>                                                                                                                                                                                        |                       |           |            | <b>32</b>    |

To emphasise the different sources and the totals

Our approach to analyzing the selected research sources proceeded as follows: (i) we extracted the criteria used to derive the tables and/or lists within each study; (ii) We inferred the nature of the challenge based on the study narrative and the type of list/table extracted; (iii) we mapped the research challenges to the phases of the three lifecycles in Fig. 2 and the framework categories in Sect. 2.1; and (iv) we created a concept-matrix indicating which studies referenced which research challenges (see worksheet "Research Challenges" in the Appendix, available online via <http://link.springer.com>).

### 3.1.2 Stage 2 – Industry Perspective

Our objective in this stage was to identify the challenges experienced by industry. Our starting point was the authors' previous work which identified industry challenges derived from interviews with 22 senior industry practitioners. The views of the regulators were derived from the four reports published after the events that motivated this study (Table 3). To identify the industry challenges in these reports, we focused on the

recommendations relevant to BPC, and by reviewing the supporting material that led to the recommendation, we were able to articulate the industry challenge. We then sourced reports from those consulting to the industry. We used <https://www.consultancy.com.au/> – an online platform for Australia's consulting industry that includes a ranking of firms operating in Australia<sup>2</sup> – to perform a filtered search of consulting firms with expertise in banking. Twelve firms were ranked in the top two tiers. We searched each of their local websites for reports, surveys and white papers focused on the challenges and root cause of compliance-related issues written between 2017-2022 (proceedings against CBA commenced in 2017 and the Royal Commission's findings were published in 2021). Of the 50 reports reviewed, we selected four to include in the analysis. The selected studies and reports are shown in Table 3.

After selecting our source documents, our approach to identify the industry challenges was as follows: i) the practitioner challenges from the authors' prior work were

<sup>2</sup> Last accessed on 30 November 2022.

**Table 2** Research perspective reference sources

| ID   | Title & references                                                                                                                                                          |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RS1  | A framework for the systematic comparison and evaluation of compliance monitoring approaches – Ly et al. (2013)                                                             |
| RS2  | A systematic literature mapping of goal and non-goal modelling methods for legal and regulatory compliance – Akhigbe et al. (2019)                                          |
| RS3  | A systematic literature review on compliance requirements management of business processes – Mustapha et al. (2020).                                                        |
| RS4  | A systematic review of goal-oriented requirements management frameworks for business process compliance – Ghanavati et al. (2010)                                           |
| RS5  | Are we done with business process compliance: state of the art and challenges ahead – Hashmi et al. (2018)                                                                  |
| RS6  | Automated compliance checking in the context of Industry 4.0: from a systematic review to an empirical fuzzy multi-criteria approach – Cunha et al. (2021)                  |
| RS7  | Automatic extraction of legal norms: evaluation of natural language processing tools – Ferraro et al. (2019)                                                                |
| RS8  | Business process regulatory compliance management solution frameworks: a comparative evaluation – El Kharbili (2012)                                                        |
| RS9  | Comparative analysis of business process modelling tools for compliance management support – Koncevics et al. (2017)                                                        |
| RS10 | Compliance evaluation featuring heat maps (CE-HM): a meta-modeling-based approach – Karagiannis et al. (2012)                                                               |
| RS11 | Compliance in e-government service engineering: state-of-the-art – Turki and Bjekovic-Obradovic (2010)                                                                      |
| RS12 | Conformance checking: a state-of-the-art literature review – Dunzer et al. (2019)                                                                                           |
| RS13 | Current research in risk-aware business process management-overview, comparison, and gap analysis – Suriadi et al. (2014)                                                   |
| RS14 | Generalizability and applicability of model-based business process compliance-checking approaches - a state-of-the-art analysis and research roadmap – Becker et al. (2012) |
| RS15 | Norms modeling constructs of business process compliance management frameworks: a conceptual evaluation – Hashmi and Governatori (2018)                                     |
| RS16 | On the formal specification of regulatory compliance: a comparative analysis – Elgammal et al. (2010)                                                                       |
| RS17 | On the fragmentation of process information: challenges, solutions, and outlook – van der Aa et al. (2015)                                                                  |
| RS18 | State-of-the-art of business process compliance approaches - a survey – Fellmann and Zasada (2016)                                                                          |
| RS19 | Time and business process management: problems, achievements, challenges (invited talk) – Eder and Franceschetti (2020)                                                     |
| RS20 | Using business process compliance approaches for compliance management with regard to digitization: evidence from a systematic literature review – Sackmann et al. (2018)   |
| RS21 | An anti-pattern-based runtime business process compliance monitoring framework – Barnawi et al. (2016)                                                                      |
| RS22 | Declarative business process modelling: principles and modelling languages – Goedertier et al. (2015)                                                                       |
| RS23 | Enabling flexibility in process-aware information systems: challenges, methods, technologies (chapter 10 business process compliance) – Reichert and Weber (2012)           |
| RS24 | On enabling integrated process compliance with semantic constraints in process management systems: requirements, challenges, solutions – Ly et al. (2012)                   |
| RS25 | Towards compliance of cross-organizational processes and their changes: research challenges and state of research – Knuplesch et al. (2012)                                 |
| RS26 | Advancing Process Audits with Process Mining: A systematic review of trends, challenges, and opportunities – Imran et al. (2023)                                            |
| RS27 | Blockchain-as-a-service for business process management: survey and challenges – Viriyasitavat et al. (2022)                                                                |
| RS28 | Compliance requirements model for collaborative business process and evaluation with process mining – González and Delgado (2021)                                           |
| RS29 | Evaluation of compliance rule languages for modelling regulatory compliance requirements – Zasada et al. (2023)                                                             |
| RS30 | Predictive compliance monitoring in process-aware information systems: State of the art, functionalities, research directions – Rinderle-Ma et al. (2023)                   |
| RS31 | Business process compliance and blockchain: How does the ethereum blockchain address challenges of business process compliance? – Meironke et al. (2019)                    |
| RS32 | Emerging challenges in information systems research for regulatory compliance management – Abdullah et al. (2010)                                                           |

mapped to the same classification framework as the research challenges in Stage 1, Step iii; ii) we extracted lists of recommendations and actions from each regulatory report, rejecting those recommendations/actions that were not directly relevant to BPC or to the banking industry and inferring the underlying industry challenge from the

residual recommendation/action by analyzing the findings leading to the recommendation/action; iv) we grouped the regulatory challenges that aligned with practitioner identified challenges and non-aligned challenges were mapped to the same classification framework in Stage 1, Step iii; iv) for the consultancy challenges, three reports were based on

**Table 3** Industry perspective reference sources

| ID               | View         | Title & references                                                                                                            |
|------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------|
| IS1              | Practitioner | This is the authors' own study that was used to develop the practitioner view.                                                |
| IS2 <sup>1</sup> | Regulatory   | Prudential inquiry into the Commonwealth Bank of Australia – APRA (2018)                                                      |
| IS3              | Regulatory   | Board governance of AML/CTF obligations at Westpac: the advisory panel review – Westpac (2020)                                |
| IS4              | Regulatory   | Royal commission into misconduct in the banking, superannuation and financial services industry (final report) – Hayne (2019) |
| IS5 <sup>2</sup> | Regulatory   | Report 594: review of selected financial services groups' compliance with the breach reporting obligation – ASIC (2018)       |
| IS6              | Consultancy  | KPMG 2021 CCO survey: sharing client perspectives on compliance imperatives – KPMG (2021)                                     |
| IS7              | Consultancy  | Accenture compliance risk study 2022 report: find calm in every compliance storm – Accenture (2022)                           |
| IS8              | Consultancy  | McKinsey: the future of operational-risk management in financial services – McKinsey & Company (2022)                         |
| IS9              | Consultancy  | Deloitte: the state of compliance survey 2021 – Deloitte (2022)                                                               |

1. APRA is the Australian Prudential Regulatory Authority. 2. ASIC is the Australian Securities & Investments Commission.

client surveys (IS6, IS7 and IS9) and we used the relevant questions in the surveys to derive the challenges, while for IS8 we used the report themes to infer the challenges; v) we combined the consultancy challenges that aligned with the practitioner/regulatory identified challenges and non-aligned challenges were mapped to the same classification framework in Stage 1, Step iii; and vi) we created a concept-matrix indicating which reports referenced which industry challenges (see worksheet “Industry Challenges” in the Online Appendix).

### 3.1.3 Stage 3 – Analyzing and Consolidating the BPCCs

Our approach to deriving a consolidated list of BPCCs was as follows:

1. Align and consolidate challenges from each perspective based on their description. While there were differences in terminology between industry and the research community, e.g., industry referred to “breaches” and “remediation”, whereas the literature referred to “violations” and “recovery”, there was a significant degree of overlap between the two perspectives.
2. Prioritize non-aligned challenges, i.e., challenges that appeared in only one of the perspectives. Challenges referred to by more than 20% of sources were retained as a BPCC. Challenges referred to by 20% or fewer sources were grouped with similar BPCCs.
3. Assign an expressive label to each BPCC and a description that adequately captures the intent of both perspectives and an industry example or quotation (Tables 4, 5, and 6).
4. Update the research and industry concept matrixes. For each concept matrix we added an extra column to create a hierarchy, entering each BPCC as a parent and

aligning the respective research and industry child challenges. For each source document, if at least one child challenge was referenced, the parent BPCC was considered to be referenced by the source (see “BPCC Source Matrix” in the Online Appendix).

5. Validate the referencing by re-analyzing each source document against the BPCC descriptions.

To calculate the relative importance of each BPCC by perspective, we first normalized the data, given there were differences in the number of studies between perspectives and the number of BPCCs within each BPC category. We divided the number of references for each BPCC by perspective by the total number of references for the perspective. To calculate the emphasis each perspective placed on a BPCC category, we divided the average number of references by perspective for each BPCC category by the total number of references for the perspective.

### 3.2 Phase 2: Identify Commercial Process Mining Solution Features and Representative Solutions

To identify suitable commercial process mining software solutions (the ‘Solutions’) with BPC relevant capabilities, e.g., conformance checking and/or compliance rule checking, we used the Everest Group PEAK Matrix Assessment (Everest Group 2022) and selected the four “leaders” (i.e., Celonis, UiPath, Software AG and Minit) plus a leading “major contender” (Apromore). Then, we extracted the features used by the Process Mining Software Comparison tool<sup>3</sup> (<https://www.processmining-software.com/>) to compare the Solutions. We also analyzed both the Gartner Market Reviews (Gartner 2022) and Everest

<sup>3</sup> This is a non-profit initiative by the Chair of Digital Industrial Service Systems at the University of Erlangen-Nürnberg. The data was last accessed on 30 November 2022.



**Table 4** BPCC summary by BPCC category (1 of 3)

| BPCC category/<br>phase                                            | BPCC descriptions with sample reference sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>C1 Compliance management – grouped by lifecycle phase</i>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Elicit                                                             | <b>C1.1 Eliciting convoluted requirements. (RS)</b> The number of process and compliance requirements has a direct bearing on the degree of difficulty for maintaining BPC [RS5]. Processes are subject to many requirements, held in different formats, in different sources, in different jurisdictions. Identifying and extracting which requirements relate to a specific process is not clear [RS2, RS4, RS7, RS11, RS29, RS30]. <b>(IS)</b> “The law is too complex” [IS4] with a myriad of legal and regulatory requirements that have expanded over time [IS3], spread over many different Acts [IS4]. “[...] it’s just wave after wave of regulation” [IS1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Formalize                                                          | <b>C1.2 Interpreting complex requirements. (RS)</b> Requirements can be ambiguous, abstract, frequently technical and/or principle based, written in natural language, using jargon, and requiring judgment to interpret [RS5, RS7, RS11, RS21, RS31]. <b>(IS)</b> Some aspects of compliance are vague and not understood [IS3] with terms such as “significant” difficult to interpret [IS5]. <b>“What does ‘we need to do the right thing’ mean?”</b> [IS1]<br><b>C1.3 Handling different requirement types. (RS)</b> There are many different types of requirements, e.g., regulatory requirements, codes of practice, contractual obligations, business policy, and standards [RS5, RS10, RS14, RS21]. There are requirements that are mandatory and requirements where the constraint can be relaxed. Some requirements attract penalties, others do not [RS16]. <b>(IS)</b> There is a distinction between requirements that are mandatory and those that are “nice to have” [IS1]. <b>“Recommendation 1.15: The law should be amended to make industry codes mandatory in financial services”</b> [IS4]<br><b>C1.4 Checking all process perspectives. (RS)</b> Temporal, data and resource requirements should also be checked, not just control-flow [RS3, RS18, RS19, RS25]. This is a particular challenge for model checking at design-time, e.g., checking deadlines for completion [RS1]. <b>(IS)</b> Not referenced<br><b>C1.5 Formalizing rules accurately.</b> There are multiple languages and approaches to formalize a rule. Graph and pattern-based languages are easier to understand but less expressive than logic-based alternatives [RS5, RS11, RS16, RS19, RS29]. <b>(IS)</b> Not referenced |
| Implement                                                          | <b>C1.6 Implementing rules correctly. (RS)</b> Integrating rules into a process can be achieved by semantic annotation, embedding the rule in the process, or using a common business vocabulary, which can be problematic [RS2, RS5]. This is further complicated by applying the same requirement across multiple processes but with different vocabulary [RS5]. Maintenance can be an issue for stored requirements given the extent of change [RS8]. Applying compliance templates may help but they are not always available [RS9, RS21, RS31]. <b>(IS)</b> Implementation guidance not captured in standard operating procedures [IS3]. <b>“Westpac did not keep a formal register to capture relevant AUSTRAC<sup>1</sup> guidance”</b> [IS3]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Analyze                                                            | <b>C1.7 Selecting the right checking approach. (RS)</b> There are multiple approaches to checking compliance that require technical expertise to understand the benefits and limitations of each [RS3, RS4, RS5, RS16, RS29]. This is complicated by the fact that the terminology varies from solution to solution, e.g., “run-time” versus “execution-time”. <b>(IS)</b> Organizations are not prioritizing how breaches are detected and may rely on reacting to losses or investigations by the regulator before taking action [IS5]. <b>“CBA has been reactive - rather than proactive and pre-emptive”</b> [IS2]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Optimize                                                           | <b>C1.8 Rationalizing rules. (RS)</b> Compliance requirements may be duplicated, overlapping, redundant or in conflict with each other [RS2, RS5, RS11, RS25, RS30, RS31, RS32]. <b>(IS)</b> Recommendation 7.3 in [IS4] requires exceptions and qualifications to be eliminated. <b>“There are layers of ‘flawed’ controls, some of which are not applied”</b> [IS1]<br><b>C1.9 Detecting requirements change. (RS)</b> Requirements change frequently; maintaining BPC is dependent on detecting and managing these changes [RS10, RS11, RS25, RS30, RS31, RS32]. <b>(IS)</b> Managing requirements change is increasingly complex as they change so frequently [IS7, IS9]. <b>“Respondents overwhelmingly identified new regulatory requirements as being one of their top challenges”</b> [IS6]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <i>C2 Business process management – grouped by lifecycle phase</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Identification                                                     | <b>C2.1 Identifying fragmented processes. (RS)</b> While there are a multitude of processes with complex interactions on a global level [RS31], the BPC literature tends to assume that processes can be readily identified, that there are Process-Aware Information Systems (PAIS) in place, and a process architecture exists [RS9, RS23, RS25]. This is not necessarily the case. <b>(IS)</b> In banking, processes are complex, fragmented, with many variants, exceptions, and workarounds [IS1]. <b>“Many of the Group’s processes are considered complex, with multiple areas involved, numerous hand offs and no one silo being responsible for the overall process”</b> [IS2]<br><b>C2.2 Extracting dispersed processes. (RS)</b> Extracting and aligning structured process data, e.g., process models, and semi-structured process data, e.g., text documents and spreadsheets, is challenging given the use of differing terminology, grammatical structures, and abstraction levels [RS17, RS31]. <b>(IS)</b> In banking, process models may be stored in multiple formats, in multiple repositories, across the enterprise. <b>“CBA’s tailoring of the model across business units ... has been compounded by a lack of documentation on how the model in each business unit works ...”</b> [IS2]                                                                                                                                                                                                                                                                                                                                                                                                        |

**Table 4** continued

| BPCC category/<br>phase | BPCC descriptions with sample reference sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Discovery               | <b>C2.3 Discovering poorly documented processes. (RS)</b> Different information is used and omitted when documenting processes depending on the user's perspective [RS17], and abstraction from low level detail is important [RS10], which may not be possible when documentation is incomplete [RS31]. Business professionals lack confidence interpreting process models and may prefer spreadsheets, verbal instructions, or written work instructions [RS17]. <b>(IS)</b> There is a lack of minimum standards and processes may not be documented [IS1, IS2]. Processes may be modeled by resources with no formal process training, and it is likely that the model only represents the "happy path", which, given the high number of process variants, is unlikely to provide an accurate representation of the actual process [IS1]. <b>"Additional complexity compounded by a lack of documentation"</b> [IS2]   |
| Analysis                | <b>C2.4 Analyzing processes with sparse data. (RS)</b> A lack of contextual information reduces the type of rules that can be checked at design time [RS21], and critical information can be missing [RS26]. <b>(IS)</b> Processes are only partially automated, resulting in a high error rate, inadequate audit trails and a lack of visibility [IS1]<br><br><b>C2.5 Analyzing processes with poor quality data. (RS)</b> Biased data can threaten audit validity [RS26] and data is inconsistent, with gaps and omissions driven by the stakeholder creating the data [RS17, RS31]. <b>(IS)</b> The lack of identifiers creates blind spots [IS1], inaccurate data are inhibiting organizations moving compliance to the Cloud [IS7] and a lack of data capability is a preventing adoption of RegTech [IS9]. <b>"We have an incomplete data management framework ... that exposes us to data quality issues"</b> [IS2] |

Each BPCC is emphasised in bold and direct quotation is in bold

Key: **(RS)** = research perspective, **(IS)** = industry perspective, examples are in bold at the end of each BPCC description

Key cont.: Characters in square brackets are sample supporting references based on the reference IDs in Tables 2 and 3

(1) AUSTRAC is Australia's financial intelligence unit and anti-money laundering and counter-terrorism financing (AML/CTF) regulator

Group PEAK Matrix Assessment (Everest Group 2022) to identify additional features.

Lastly, an evaluation of the Solutions had been undertaken in 2020 by the team responsible for the Process Mining Software Comparison tool. We extracted the evaluation data and used this as the starting point of our analysis. As this data related to an evaluation completed in 2020, and each of the Solutions has been through a series of upgrades since that time, we completed a desktop review of the Solutions based on publicly available information. We checked and updated information based on our testing of trial versions of the Solutions, and associated support materials, e.g., online help, community forums, and FAQs where relevant. Subsequently, we assessed the current capabilities of the Solutions relative to the identified features. For example, the ability to *Import File Types* was identified as a feature and the Solutions demonstrated differing capabilities with regards to the range of file types they supported.

### 3.3 Phase 3: Map the Derived Process Mining Features and Capabilities to the BPCCs

In this phase, we mapped the features and capabilities identified in phase two that may address the BPCCs identified in phase one. The mapping was based on the expert opinion of the research team. We applied three ratings: "Satisfied", "Did Not Satisfy" and "Enabled". A "Satisfied" rating refers to cases where at least one of the

Solutions had one or more features that addressed the BPCC. "Did Not Satisfy" refers to cases where none of the Solutions addressed the BPCC. The "Enabled" rating was restricted to the *Organizational* BPCC category, as a software solution would not be able to satisfy the management issues in this BPCC category. Where the features of the Solutions could potentially help management address these issues, we applied an "Enabled" rating. We summarized the results in a BPCC/feature matrix describing, where relevant, how the features satisfied each BPCC.

## 4 Results

In this section, we first outline the BPCCs we identified from both a research and industry perspective, followed by the solution features, and an assessment of the capabilities of the five solutions relative to these features. Finally, we table the results of the mapping between features and the BPCCs.

### 4.1 BPCCs Identified

Stage 1 identified 65 research challenges and Stage 2 identified 36 industry challenges. Following consolidation there were 41 BPCCs. The mapping of the industry and research challenges to the BPCCs is presented in the "BPCC Mapping" worksheet in the Online Appendix. We describe the BPCCs in Tables 4, 5 and 6. The tables are

**Table 5** BPCC summary by BPCC category (2 of 3)

| BPCC category/<br>phase                                                  | BPCC descriptions with sample reference sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>C2 Business process management cont. – grouped by lifecycle phase</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Redesign                                                                 | <b>C2.6 Working with diverse modeling languages. (RS)</b> The research community acknowledges that there are a range of process modeling languages in use, e.g., BPMN, Petri nets [RS3, RS5, RS31], spreadsheets and text-based work instructions [RS17]. <b>(IS)</b> The tools used for documentation are inadequate [IS1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Implementation                                                           | <b>C2.7 Implementing complex processes. (RS)</b> There is an assumption that a PAIS or Business Process Management System is in place that will support implementation [RS23, RS28] and exceptions can create noise [RS26] in a complex environment [RS31]. <b>(IS)</b> Banking IT ecosystems are complex, driven by a proliferation of technologies, the rate of change and a multitude of IT providers [IS3]. The technology is inflexible, disconnected [IS1] and requires “heavy” continuing investment [IS3]. Even a small change to a compliance requirement may take many months to implement [IS1]. <b>“The age, complexity and diversity of products and business unit systems inhibit the identification of incidents”</b> [IS5].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Monitoring                                                               | <b>C2.8 Monitoring shortcomings. (RS)</b> Not all processes are monitored, and manual audit is the only option [RS26]. Monitoring in the BPC literature is largely associated with checking compliance at run time and there is an implicit assumption that the relevant process data are available [RS21, RS31, RS32]. <b>(IS)</b> There is a need to move from qualitative, manual controls to data-driven, real-time monitoring [IS8]. While systems and their queues may be monitored, it is often difficult to capture the process view [IS1]. The reliance on manual, sample-based compliance checking means errors can go undetected for long periods of time [IS4]. <b>“The major financial groups took an average of 1,726 days ... to identify an incident”</b> [IS5].<br><br><b>C2.9 Detecting process change. (RS)</b> Processes change frequently. Maintaining BPC is dependent on detecting and managing these changes [RS5, RS10, RS24, RS30, RS31]. <b>(IS)</b> Poor change management has led to failure [IS3]. It is not just the amount of change; it is the pace of change that makes it challenging [IS8]. <b>“The scope and speed of enterprise transformations put additional stress on the compliance function”</b> [IS8].           |
| <i>C3 Compliance checking – grouped by lifecycle phase</i>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Design-time                                                              | <b>C3.1 Checking at design time. (RS)</b> Model checking is complex and is neither able to check all process perspectives, for example time-based requirements such as the duration between two tasks, nor can it anticipate context-specific change at execution time, e.g., a system outage may require dynamic changes to the process execution [RS21, RS24, RS31]. The “state space explosion problem constitutes a big obstacle for the practical use...of model checking”, with larges sets of rules and certain aspects of process behavior, such as infinite loops, model verification may not be possible [RS5, RS8, RS23, RS25]. <b>(IS)</b> Not referenced.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Run-time                                                                 | <b>C3.2 Checking at run time. (RS)</b> The challenges of run-time monitoring are dependent on whether it is a forward-looking or backward-looking approach [RS20] and is complicated by the range of approaches to capture events [RS1, RS21], and the granularity of the data [RS21, RS30]. It must support activity lifecycle checking that describe the allowed executions of events associated with non-atomic activities [RS1, RS30]. Furthermore, not all requirements can be checked, e.g., those that use technical devices for encryption [RS21]. <b>(IS)</b> Not referenced.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Audit                                                                    | <b>C3.3 Audit checking (RS)</b> is traditionally a manual and sample-based activity, which is expensive and subject to error [RS26]. It is also a reactive approach [RS5, RS8]. By the time a violation is detected it is often too late to address the issue [RS24]. <b>(IS)</b> While the risk function is “discarding the ‘rearview mirror’ approach” [IS8] and becoming more proactive [IS8] there is still a reliance on detective controls [IS1]. <b>“72% of organizations surveyed still rely on audit results for assessing the effectiveness of their compliance program”</b> [IS6].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <i>C4 Compliance analysis</i>                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Reporting                                                                | <b>C4.1 Measuring what matters. (RS)</b> Detailed violation metrics are regarded as a key BPC requirement [RS10, RS12, RS20] but, with a range of metrics, e.g., ‘fuzzy’ metrics, measuring compliance and conformance is not always straightforward [RS1, RS12, RS31]. <b>(IS)</b> Measurement remains difficult and there is a need to create metrics that ‘connect the data dots’ [IS8], this includes linking metrics, e.g., operational and behavioral metrics [IS6]. In [IS5] there were 29 instances where the start of the breach (violation) was not identified. <b>“Recommendation 12: Gaps in reporting and metrics hampered the effectiveness of the Board”</b> [IS2].<br><br><b>C4.2 Aligning reporting (RS)</b> Dashboards and reports must be able to aggregate and visualize results in different formats and support multiple perspectives [RS1, RS5, RS9, RS10, RS21, RS30]. <b>(IS)</b> Getting the balance right between summarized and detailed reporting hampered the Board at CBA [IS2]. Information is recorded in a fragmented fashion, not searchable, missing key data making it difficult for staff to ‘connect the dots’ [IS5] <b>“Management did not know and hence could not inform the Board until they did know”</b> [IS3]. |

**Table 5** continued

| BPCC category/<br>phase                               | BPCC descriptions with sample reference sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>C5 Compliance enactment (violation management)</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Recover                                               | <p><b>C5.1 Creating timely alerts. (RS)</b> Users should be notified of violations in a meaningful way [RS5, RS8, RS21, RS24, RS30]. <b>(IS)</b> The “alarm bells“, may come from a range of sources not just the process management system, e.g., customer complaints or whistleblowers [IS2]. One in seven institutions failed to notify the regulator within the time allowed [IS5]. <b>“The Board did not receive alerts on either incidents or themes”</b> [IS2].</p> <p><b>C5.2 Handling multiple violations. (RS)</b> Recording, maintaining and reporting accurate violation information is required to support traceability [RS24, RS25] but creating evidence of compliance can be difficult [RS32]. After an initial violation, a solution must continue to check for further violations, particularly at run time with multiple, concurrent instances [RS1, RS30]. <b>(IS)</b> The industry must get better at record keeping and documenting the results of violations [IS5]. <b>“There has been continued weakness in identifying larger emerging and systemic issues“</b> [IS2].</p> <p><b>C5.3 Prioritizing the right violations. (R)</b> As discussed in C1.3 some compliance rules can be relaxed and others trigger compensation, which requires rule prioritization [RS21, RS23, RS24, RS30]. <b>(I)</b> Monitoring different degrees of compliance is challenging [IS2]. <b>“Westpac should have implemented more robust monitoring of their transactions for certain types of behaviors earlier than it did”</b> [IS5].</p> <p><b>C5.4 Finding root cause. (RS)</b> A solution must be able to identify the point of failure in the process and the root cause, providing intelligible user feedback [RS1, RS5, RS8, RS11, RS21, RS24, RS25] but there can be insufficient traceability [RS31]. <b>(IS)</b> The capability to undertake root cause analysis is immature [IS2] and late detection makes it more difficult to identify the root cause [IS5]. <b>“Recommendation 12: Root causes and not merely issues are addressed in a timely and effective manner“</b> [IS2].</p> |

Each BPCC is emphasised in bold and direct quotation is in bold

Key: **(RS)** = research perspective, **(IS)** = industry perspective, examples are in bold at the end of each BPCC description

Key cont.: Characters in square brackets are sample supporting references based on the reference IDs in Tables 2 and 3

aligned by BPCC category and the phases/elements within each category are based on the discussion in Sect. 2.1. We present a view of each BPCC from both the research and industry perspective, as well as an example solely from the industry perspective.

We ranked each BPCC by perspective based on its number of references – where two or more BPCCs had the same number of references, they were given the highest rank for that set of values. There were no overlapping BPCCs when considering the five highest ranked BPCCs from each perspective. 42% of industry challenges were in the *C1 Compliance management* and *C3 Compliance checking* categories, whereas only 22% of industry challenges were in these categories. This difference in emphasis is further highlighted in Fig. 3a. The frequently referenced research BPCCs were more technical in nature, e.g., *C1.5 Formalizing rules accurately*, *C1.7 Selecting the right checking approach*, while the frequently referenced industry BPCCs were more practical, and process focused, e.g., *C2.8 Monitoring shortcomings*, *C6.3 Scaling across the enterprise*, and those BPCCs assigned to the *C7 Organizational* category. When viewed at a BPCC category level, the differences between the perspectives are clearer (Fig. 3b).

## 4.2 Commercial Process Mining Solution Features

We extracted 55 features from the Process Mining Software Comparison tool. We aligned the features to a modified version of the categories used in the comparison tool by consolidating the operational support category with the deployment, security and permissions category to form *F7 Operational support*. We then reassigned *F4.10 Alert generation* to the *F4 Conformance checking* category and both *F5.1 Predictive analytics* and *F5.2 Prescriptive analytics* to the *F5 Advanced enhancement capabilities* category. Our review of Gartner Market Reviews (Gartner 2022) and Everest Group PEAK Matrix Assessment (Everest Group 2022) led to an additional eight BPC-specific features, marked with an asterisk in Fig. 4: *F4.3 Compliance rule checking*, *F4.4 Compliance rule templates*, *F4.11 Real-time monitoring*, *F5.4 Task mining*, *F5.5 Execution support*, *F6.13 Scalability*, *F7.6 Training & support*, and *F7.7 Affiliated partnerships*. Figure 4 provides a summary of the features and their categories. A description of the features is presented in the “Features“ worksheet in the Online Appendix and the assessment of the Solutions is on the “Feature Rating by Vendor“ worksheet in the same file.

**Table 6** Challenge summary by BPCC category (3 of 3)

| BPCC category/<br>phase                                     | BPCC descriptions with sample reference sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>C5 Compliance enactment (violation management) cont.</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Recover                                                     | <b>C5.5 Recovering satisfactorily from violations (RS)</b> After a violation has occurred a solution must support the user define a corrective course of action, such as change the process or trigger compensation [RS1, RS5, RS8, RS21, RS30]. <b>(IS)</b> Poor project execution hampers issue closure and some issues are closed without resolving the root cause [IS2]. Remediation is still a significant cost of violation and organizations have increased the focus on investigations [IS6]. On average, investigating an incident takes 150 days before it is reported [IS5]. <b>"One example is ... compensating customers affected by failures that were first identified in 2003, too many of which were still far from complete ... in March 2018"</b> [IS4]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Prevent                                                     | <b>C5.6 Predicting violations accurately. (RS)</b> Based on analysis of the steps already executed in a process, this challenge is focused on predicting the likelihood of future violations, e.g., the likelihood of missing a deadline [RS1, RS5, RS21, RS30]. It is particularly challenging for processes that: i) change frequently, ii) are non-repetitive, iii) are new, iv) have many variants, and v) have few instances [RS19]. <b>(IS)</b> Organizations recognize that they need to invest in predictive analytics [IS6], and that proactive identification of issues is weak [IS2]<br><br><b>C5.7 Preventing repeat violations. (RS)</b> This challenge involves predicting when a violation is about to occur and either recommending or executing the "next best task" to prevent the violation [RS1, RS5, RS8, RS20]. The ability to provide users with an impact assessment of options at different levels of granularity is challenging [RS30]. <b>(IS)</b> Regulators would like to see a more proactive approach [IS2], but organizations are not learning from their failures and/or sharing opportunities to prevent violations [IS5]. <b>"Licensees were too often reactive ... neglecting the lessons learned opportunity"</b> [IS5]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <i>C6 Business support</i>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| User                                                        | <b>C6.1 Supporting the user. (RS)</b> Many formal languages are complex and frustrating for non-technical users where comprehensibility is an issue and there is limited support [RS6, RS9, RS16, RS22, RS25, RS31, RS32]. <b>(IS)</b> There is a need to provide compliance systems' training [IS5]. In global organizations, multi-lingual support is also critical, and compliance policies must be easy to find and interpret at all levels [IS6]. <b>"Are compliance systems and processes user friendly?"</b> [IS5]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Technical                                                   | <b>C6.2 Integrating securely. (RS)</b> BPC data are highly sensitive, and the systems architecture is complex, with cross-organizational requirements and different deployment models presenting integration challenges [RS3, RS9, RS25, RS26]. The implementation of tools to support BPC often leads to dependence on the corresponding service provider in terms of financial aspects, data security and data privacy [RS31]. <b>(IS)</b> There are multiple operating environments and many applications that must be integrated securely in a high-risk domain [IS1]. <b>"The one [roadblock] that springs to mind straightaway is integration. Our technology architecture is far from simple. There's bits and pieces logged [sic] all over the place."</b> [IS1]<br><br><b>C6.3 Scaling across the enterprise. (RS)</b> In practice, modeling cross-organizational compliance rules, propagating and managing change efficiently and allowing for growth is challenging when executing enterprise-wide BPC [R10, RS25]. <b>(IS)</b> Banking is an extreme example of the scaling problem [IS1] Risk processes must be integrated and aligned across the enterprise [IS7] and only 2% have fully automated their operational risk (and compliance) management [IS9]. <b>"No systems for processing the numbers of transactions or variety of arrangements offered by banks will ever operate perfectly"</b> [IS4]<br><br><b>C6.4 Demonstrating real-world use. (RS)</b> The importance of demonstrating real-world application with prototypes is recognized [RS13, RS14] and many contributions include a technical artifact [RS18], however there is a significant time difference between testing a real-world scenario and commercializing a solution. <b>(IS)</b> Not referenced |
| <i>C7 Organizational</i>                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                             | <b>C7.1 Addressing business model complexity. (RS)</b> Managing the "interplay of change and compliance in the context of cross-organizational processes" is challenging [RS25] particularly with multiple partners and organizational structures in a multi-jurisdictional context [RS5, RS27, RS31]. <b>(IS)</b> Operational complexity has increased [IS8], and compliance is struggling to keep pace with the business model [IS7]. Multi-brand and matrix managed structures impair end-to-end visibility and ownership [IS3]. <b>"Errors occur when the left hand does not know what the right hand is doing"</b> [IS4]<br><br><b>C7.2 Implementing enterprise-wide governance. (RS)</b> There is a lack of communication, the approach to is ad hoc and a lack of standard practices and rigor [RS31, RS32]. <b>(IS)</b> It is challenging to keep compliance requirements, business processes and the risk management framework aligned [IS2]. <b>Only 52% of respondents thought that the compliance risk management framework was highly integrated with business and first line risk frameworks</b> [IS9]<br><br><b>C7.3 Changing an indifferent culture. (RS)</b> BPC is seen as a bureaucratic burden, not as a "value add" [RS31, RS32] with few management role models [RS31]. <b>(IS)</b> Risk management is dominated by a 'tick-the-box' process mentality [IS2]. Only 15% of respondents in [IS9] agree that compliance culture is fully aligned across the organization and embedding cultural change requires leaders to role model appropriate risk behaviors [IS2]. <b>"Obvious priorities will include driving cultural change"</b> [IS3]                                                                                                                            |



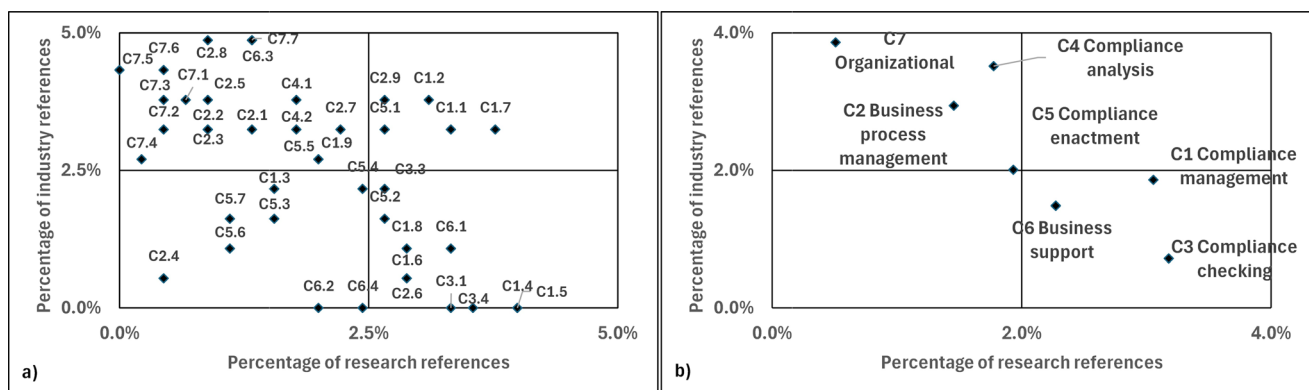
**Table 6** continued

| BPCC category/<br>phase | BPCC descriptions with sample reference sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p><b>C7.4 Resolving conflicting objectives.</b> (RS) There are conflicts of interest between stakeholders [RS31]. (IS) There are differences between the ‘voice of the customer’, the ‘voice of finance’ and the ‘voice of compliance’ [IS2]. <b>“We acknowledge that ... licensees have many, often competing, priorities within their businesses”</b> [IS5]</p> <p><b>C7.5 Clarifying accountability.</b> (RS) Not referenced. (IS) There is a lack of top management support [RS32], unclear accountability in the Three Lines of Defense model [IS3] and consequence management is not always consistently applied [IS5]. <b>“Processing errors arose from ... a(n) absence of end-to-end accountability”</b> [IS4]</p> <p><b>C7.6 Building staff knowledge.</b> (RS) BPC requires legal, compliance and business process expertise and there are knowledge gaps and errors [RS31, RS32] and a lack of resources [RS32]. (IS) It is hard for organizations to find qualified resources [RS32], compliance functions are immature [IS2], and operational teams lack specialist expertise [IS3].</p> <p><b>Recommendation 13: CBA build up the capabilities and subject matter expertise of operational and compliance risk staff...</b> [IS2]</p> <p><b>C7.7 Creating economic transparency.</b> (RS) BPC is expensive, but costs, benefits, and returns are difficult to quantify [RS10, RS19, RS22, RS31]. (IS) Investment is reactive and short-sighted [IS1]. Organizations only invest where it is mandatory [IS2]. <b>“90% of compliance leaders expect... compliance-related operating costs [to increase] by up to 30%”</b> [IS7]</p> |

Each BPCC is emphasised in bold and direct quotation is in bold

Key: (RS) = research perspective, (IS) = industry perspective, examples are in bold at the end of each BPCC description

Key cont.: Characters in square brackets are sample supporting references based on the reference IDs in Tables 2 and 3



**Fig. 3** BPCC and BPCC category references by source: industry versus research

#### 4.3 Process Mining Features Mapped to BPCCs

The solutions “Satisfied” 27 Of the 34 non-*Organizational* BPCCs and “Enabled” all *Organizational* BPCCs. Of the seven BPCCs that were not “Satisfied”, five are essentially out-of-scope for current commercial process mining solutions: *C1.1 Eliciting convoluted requirements*, *C1.2 Interpreting complex requirements*, *C1.8 Rationalizing rules*, *C1.9 Detecting requirements change*, and *C3.1 Checking at design time*. The complete mapping of the features to the BPCCs is in the “BPCC vs. Feature Concept Matrix” of the Online Appendix.

The intent of our analysis was to determine whether a feature or combination of features could be applied that would satisfy the challenge and we summarize our findings

in Tables 7 and 8. However, the flexibility of process mining as a platform does not rule out the viability of options other than the ones presented here. It is also important to note that we have only described the most salient features that would satisfy a specific BPCC, as some features, particularly in the *F1 Data management* category would apply to all of the “Satisfied” BPCCs.

One noticeable aspect of the mapping is that 19 BPCCs are “Satisfied” by more than five features each. Even after excluding the *F1 Data management* features, which accounted for the highest number of features (equal with *F4 Conformance checking*), 15 BPCCs are “Satisfied” by more than five features each. In some cases, this is indicative of the range of options that a user may need to configure to address a BPCC, e.g., *C1.7 Selecting the right*

| F1 Data management                         |                             | F2 Process graph                  |                                |
|--------------------------------------------|-----------------------------|-----------------------------------|--------------------------------|
| F1.1 Import Files Types                    | F1.7 Character Encodings    | F2.1 As-is process visualization  | F2.6 Graph abstraction         |
| F1.2 Database Connections                  | F1.8 Attribute Types        | F2.2 Export as-is process graph   | F2.7 Frequency metrics         |
| F1.3 Adapters/Connectors                   | F1.9 Specify Business Hours | F2.3 Performance highlighting     | F2.8 Time metrics              |
| F1.4 Integrated ETL Functionality          | F1.10 Define Event Order    | F2.4 Process animation (replay)   | F2.9 Additional graph metrics  |
| F1.5 Data Anonym. & Pseudonym.             | F1.11 Start/End Timestamp   | F2.5 Search and filter in graph   |                                |
| F1.6 Data Loading                          |                             |                                   |                                |
| F3 Process analysis and analytics          |                             | F4 Conformance checking           |                                |
| F3.1 Process benchmarking                  | F3.5 Case and activity list | F4.1 Compare as-is and target     | F4.7 Four-eyes principle       |
| F3.2 Process benchmarking (different logs) | F3.6 View case details      | F4.2 Target model creation        | F4.8 Sequence filtering        |
| F3.3 Root cause analysis                   | F3.7 Rework analysis        | F4.3 Compliance rule checking*    | F4.9 Conf. root cause analysis |
| F3.4 Variant breakdown                     | F3.8 Transition details     | F4.4 Compliance rule templates*   | F4.10 Alert generation         |
|                                            |                             | F4.5 In-graph conf. visualisation | F4.11 Real-time monitoring*    |
|                                            |                             | F4.6 List of conf. violations     |                                |
| F5 Advanced enhancement capabilities       |                             | F6 View, monitoring and reporting |                                |
| F5.1 Predictive analytics                  | F5.5 Decision rule mining   | F6.1 Export reports               | F6.6 Advanced charts           |
| F5.2 Prescriptive analytics                | F5.6 Task mining*           | F6.2 Export charts and tables     | F6.7 World map                 |
| F5.3 Organizational mining                 | F5.7 Execution support*     | F6.3 Custom dashboards            | F6.8 Save filter settings      |
| F5.4 Scenario simulation                   |                             | F6.4 Custom metrics/KPIs          | F6.9 UI languages              |
|                                            |                             | F6.5 KPI thresholds               | F6.10 Share & collaborate      |
| F7 Operational support                     |                             |                                   |                                |
| F7.1 Deployment model                      | F7.3 Role-based access      | F7.5 Audit logs                   | F7.7 Affiliated partnerships*  |
| F7.2 Scalability*                          | F7.4 User authentication    | F7.6 Training & support*          |                                |

**Fig. 4** Commercial process mining solution features by feature category

*approach*, in other cases, satisfying a BPCC may require a combination of features, e.g., *C1.4 Checking all process perspectives*.

## 5 Discussion

Our answer to RQ1 suggests that while both the research community and industry identify many of the same challenges, there are significant differences in the relative importance each group attributes to them. The research community is predominantly focused on developing languages to interpret and formalize compliance requirements, and to check those requirements at design time and run time, in line with the findings in Sackmann et al. (2018). By contrast, industry stakeholders acknowledge a need to prioritize a more proactive approach, but the process and systems challenges are more important and would appear to preclude design-time checking, and limit run-time checking.

The processes referred to in the regulators' reports are mature, therefore "design time" is concerned with enhancing and improving existing processes, not designing "new to bank" processes. Process models, if they exist for these processes, are unlikely to accurately describe their behavior. At run time, except for time-critical controls, e.g., fraud monitoring, it is unlikely that a bank would want to add to the overall complexity of its systems environment by introducing non-time-critical, real-time monitoring.

These issues are compounded by a range of organizational challenges that impede banks from prioritizing a more proactive approach and are largely beyond the scope of the BPC research community. Therefore, the support that the research community could provide to industry would appear to be limited.

In terms of RQ2, seven BPCCs were not "Satisfied", three of which (C1.1, C2.2 and C3.1) relate to phases in the compliance management lifecycle and, therefore, are critical to a comprehensive BPC solution. Moreover, of the BPCCs that were "Satisfied", many required multiple features, e.g., more than five features were required to satisfy 19 BPCCs. As a result, a BPC solution based on process mining functionality may be confusing and 'clunky' for users.

With both research and commercial process mining limitations, a comprehensive, credible, 'fit for purpose' platform for a BPC solution would appear to be, in the short term at least, out of reach. However, by changing the paradigm, our subsequent analysis identified workaround options to address both the research and process mining limitations. We discuss each of the BPCCs that were rated as "Did not satisfy" in Phase 3.3 below. We then discuss our concerns with checking non-time-critical compliance requirements at run time, and the complexity faced by users when selecting the right checking approach (C1.7). Finally, we discuss the impact of multiple features being required to satisfy a BPCC.

## 5.1 Addressing the Limitations

- C1.1 *Eliciting convoluted requirements*: While the solutions have robust ETL capabilities, the essence of this BPCC is the difficulty banks have in identifying the relevant compliance requirements – a challenge well known to the research community. However, in practice, Principle 2 of the Basel Committee's<sup>4</sup> “Revisions to the Principles for the Sound Management of Operational Risk” (Basel Committee on Banking Supervision 2021), requires banks to “develop, implement and maintain an operational risk management framework (ORMF) that is fully integrated into the bank's overall risk management processes”. A requirement of the ORMF is to provide an “inventory [of the] risks and controls implemented by all business units (e.g., in a control library).” In essence, applying the *F1 Data management* features to extract and import into or connect to an existing “inventory” of risks and controls, i.e., a risk, obligations and control register (ROCR) to the process mining platform, from a single ORMF source, is a workaround to address C1.1. This approach is dependent on a complete ROCR and an ability to store the ROCR. It will not identify risk and control gaps. In this case, *F5.5 Decision rule mining* may be one option to derive the rules from the control flow, automatically discovered from the event log.
- C1.2 *Interpreting complex requirements*: As stated above, the ORMF requires banks to maintain a ROCR. The ORMF also requires the banks to: “Describe the bank's approach to ensure controls are designed, implemented and operating effectively” and to “Describe the bank's approach to establishing and monitoring thresholds or limits for inherent and residual risk exposure”. These descriptions serve as a proxy for a natural language interpretation of the compliance requirement, bypassing the challenge of *C1.2 Interpreting complex requirements*. With a clear statement of a compliance requirement, i.e., the control description, compliance rule templates (F4.4) can be applied to formalize the requirement in a standard and consistent way. While this approach is suitable for testing the operating effectiveness of controls, it is limited in its ability to detect control design defects as it assumes the choice of control and how it is to be implemented are reflected accurately in the control description.

- C1.8 *Rationalizing rules*: Formalizing and instantiating compliance rules based on compliance rule templates (F4.4) and being able to visualize the rules in a process model (functionality that would need to be developed but similar to F4.5), would make it easier to identify potential conflicts and redundancies. Additionally, a customized dashboard (F6.3) that would allow the user to list all controls and compliance rules associated with a process, along with the definitions of instantiated rules, i.e., saved filter settings (F6.8), would also make it easier to identify potential redundancy and conflict. These workaround options do not automate the task of identifying rules to be rationalized, but they do support the user to do so.
- C1.9 *Detecting requirements change*: Similar to C1.8, while process mining does not satisfy this challenge, a file comparison of the current risk, obligations and controls within the platform and the new version of the register can highlight changes to support the user identify which risk, obligations or controls have changed and then propagate updates to the relevant compliance rules. This feature does not exist within solutions we tested; hence we have retained a rating of ‘Did Not Satisfy.’
- C3.1 *Checking at design time*: Process mining does not support checking or verifying a model against a list of requirements. As such, for a “new to bank” process, modeled for the first time, process mining cannot check the model. However, for existing, mature processes undergoing re-design, there are a number of workaround options to satisfy this BPCC:
- i. An event log can be generated by simulation (F5.4), deriving its distribution parameters from the existing process prior to modification, and compliance rule checking (F4.3) can be applied to the simulated event log.
  - ii. For a “new to bank” process, where execution data does not exist, a simulation approach similar to item i. can also be applied once testing cycles begin and test logs are generated.
  - iii. In very limited circumstances, potentially at the early stages of design, while compliance requirements are still being formalized, industry benchmarking models (F3.1, F3.2) can act as a proxy for a target model, e.g., comparing the number of resources assigned to a specific activity can be a proxy for a high-risk activity.

<sup>4</sup> The Basel Committee on Banking Supervision (BCBS) is the primary global standard setter for the prudential regulation of internationally active banks.

- C5.3** *Prioritizing the right violations:* Determining the significance of violations is a function of many factors, for example, (i) the type of requirement, e.g., a regulatory requirement versus a policy requirement; (ii) the root cause of the violation, e.g., human error vs. deliberate design; (iii) the frequency of violation; (iv) the impact of the violation, e.g., a “near miss” versus large financial loss; and (v) the remediation progress, e.g., speed of identification, acknowledgment of violation, “make good” provisions executed. Some of these can be supported by a process mining platform. For example: (i) imported controls can be assigned with a prioritization category attribute, based on the potential consequences of violation; (ii) root cause categories can be assigned reason code attributes, which can be associated with prioritization to form the basis for custom metrics (F6.4) with tolerance thresholds (F6.5); and (iii) elapsed time can be derived (F2.8) based on when the violation triggering event happened and the alert being generated (4.10). However, outside of the simplest violations, e.g., an SLA not being met, these features will contribute to determining the prioritization but will require supplemental information such as progress on the remediation schedule.
- C5.5** *Recovering satisfactorily from violations:* There are two aspects to consider with this BPCC. For processes that are still executing, using compliance rule templates (F4.4) to design-in compensatory or recovery rules when a control is being formalized will satisfy the BPCC in this context. For example, a control to contact a customer within four hours may have a compensatory control such as: “Unless the customer has requested a specific call back time”. The second aspect refers to instances where violations have been detected post execution, which should trigger a separate recovery process. Treating the event log generated by the recovery process as a process to be monitored and checked like other processes, e.g., with controls to monitor stage gate progress, would at least highlight compliance with the recovery process’ compliance requirements.

Satisfying *C3.2 Checking at run time* was largely enabled by *F4.11 Real-time monitoring*. Given the systems challenges described in *C2.7 Implementing complex processes*, a near real-time approach may be preferable, where event logs are updated (F1.6) at regular intervals, e.g., every 10–15 min. This approach mitigates the operational performance risk of monitoring a large number of compliance requirements in the ‘live’ environment and avoids the

technical challenge of monitoring event streams. While there are requirements that banks must monitor in real time, e.g., for fraud monitoring or sanctions screening, large, mature banks already deploy a wide range of bespoke applications to perform these tasks. Therefore, this risk is also mitigated. Near real-time monitoring also simplifies the end-to-end BPC approach as event logs are used as a consistent base representation of the process for *C3.1 Checking at run time*, *C3.1 Checking at design time*, and *C3.3 Audit checking*.

There is also the issue of simplifying BPC for a business user. With so many different BPC approaches available, some highly technical, the business user faces the difficult task of *C1.7 Selecting the right checking approach*, with little guidance (Zasada et al. 2023). In the approach presented here: (i) a standard ROCR serves as the single source of requirements; (ii) the requirements are formalized consistently by applying a comprehensive set of rule checking templates; (iii) the requirements are instantiated as compliance rules on an event log, i.e., a standard representation of the process; (iv) the compliance rules can then be checked consistently across all phases of the BPC lifecycle. *C1.7 Selecting the right checking approach* is significantly simplified for the user – it is a First-Order, Linear Temporal Logic over Finite Traces approach, irrespective of which phase of the BPC lifecycle is being checked. The only variation for the user is to select an appropriate event log, representative of the phase.

We have summarized the impact of this paradigm change in Table 9. The “Assessment” columns states the initial Sect. 4.3 rating, followed by a “Paradigm Shift Satisfied” rating for those BPCCs in Sect. 5.1 that were subsequently addressed. With only one “Did Not Satisfy” rating, our analysis demonstrates that process mining solutions can serve as a platform for a comprehensive, easier-to-use, end-to-end BPC solution. It does not address the fact that multiple features may be required to satisfy a BPCC, which may make the user experience feel “clunky”. To address this requires a dedicated user interface, designed with the requisite, underlying features running in the background. Designing such a framework is the next stage of our research.

## 5.2 Implications

In broad terms, the implications of this study are that research efforts need to be more aligned to industry, vendors must develop a comprehensive, easy-to-use BPC solution, and industry stakeholders must address the foundational issues at the root of the BPCCs. We discuss the detailed implications below.

*Implications for researchers:* In a banking context, researchers must recognize that industry is more concerned



**Table 7** Mapping features to BPCCs by BPCC category (1 of 2)

| BPCC category                                                      | BPCCs and the features that satisfy them                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>C1 Compliance management – grouped by lifecycle phase</i>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Elicit                                                             | <b>C1.1 Eliciting convoluted requirements.</b> Did not satisfy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Formalize                                                          | <b>C1.2 Interpreting complex requirements.</b> Did not satisfy.<br><b>C1.3 Handling different requirement types.</b> This BPCC can be satisfied by event log querying (F4.3), which will check all types of requirements (see C1.5 below). Attribute data would be required to distinguish between mandatory/non-mandatory types or where compensation must be met.<br><b>C1.4 Handling all process perspectives.</b> This BPCC is satisfied by a range of features including time metrics (F2.8), transition details (F3.8) and sequencing filtering, i.e., control flow (F4.8) as well as specific resource metrics (F4.7) and other metrics (F2.7, F2.9, F3.7). Additional insight on each of the perspectives can be provided by including task-mining data (F5.6) and robotic process automation (RPA) data (van der Aalst 2020) from partnerships (F7.7) and organizational mining (F5.3) to complement the resource perspective.<br><b>C1.5 Formalizing rules accurately.</b> This can be satisfied by applying compliance rule checking and templates (F4.3, F4.4) to formalize compliance requirements. |
| Implement                                                          | <b>C1.6 Implementing rules correctly.</b> This is satisfied by instantiating rules using fields/attribute data in the event log as filter parameters (F1.8), built-in conformance checking rules (F4.7, F4.8), compliance rule templates (F4.4) and saving filter settings (F6.8).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Analyze                                                            | <b>C1.7 Selecting the right checking approach.</b> A platform-based approach satisfies the challenge by presenting the different approaches with the same “look and feel”, irrespective of the BPC management lifecycle phase. However, process mining does not support design-time checking and to check compliance at run time or post execution requires a wide range of features: i) the ability to use benchmarking templates as a prescribed model proxy (F3.1) or variant analysis to check a variant of a log with no violations to a variant with violations (F3.4), and ii) a range of conformance and compliance checking features to define a target model (F4.2), compare it to the “As-Is” (F4.1), check at run time (F4.11) check and re-use compliance rules (F4.3, F4.4, F4.7, F4.8, F6.8) to highlight violations (F4.5 and F4.6).                                                                                                                                                                                                                                                             |
| Optimize                                                           | <b>C1.8 Rationalizing rules.</b> Did not satisfy.<br><b>C1.9 Detecting requirements change.</b> Did not satisfy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <i>C2 Business process management – grouped by lifecycle phase</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Identification                                                     | <b>C2.1 Identifying fragmented processes.</b> This can be satisfied by applying a range of the <i>F1 Data management</i> extract, transform and load (ETL) features (F1.1-F1.8, F1.11) to generate an event log (F2.1, F2.6).<br><b>C2.2 Extracting dispersed processes.</b> This can be satisfied by applying a range of the <i>F1 Data management</i> ETL features (F1.1-F1.4, F1.6-F1.7) to extract an event log or process model from across multiple repositories.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Discovery                                                          | <b>C2.3 Discovering poorly documented processes.</b> A process model that consistently and accurately represents the behavior of the underlying process can be discovered automatically (F2.1), exported for other stakeholders to use (F2.2), and abstracted to a level applicable to the stakeholder (F2.6). Alternatively, a model can be created manually in BPMN (F4.2).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Analysis                                                           | <b>C2.4 Analyzing processes with sparse data.</b> Where processes are partially automated, the ability to automatically discover a process (F2.1) based on available data and graph abstraction (F2.6), as well as performance highlighting (F2.3) and a range of metrics (F2.7-F2.9) can help users visualize the automation gaps, e.g., by highlighting the duration between two activities that are separated by a known manual activity.<br><b>C2.5 Analyzing processes with poor quality data.</b> This can be satisfied by applying a range of the <i>F1 Data management</i> features to improve data quality prior to loading. (F1.1-F1.7).                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Redesign                                                           | <b>C2.6 Working with diverse modeling languages.</b> This can be satisfied by discovering process models automatically and exporting them (F2.1, F2.2), along with the ability to generate a target process model (F4.2), ensuring that a consistent modeling standard is applied.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Implementation                                                     | <b>C2.7 Implementing complex processes.</b> This can be satisfied by improving transparency of the automatically discovered As-Is process (F2.1), at different levels of abstraction (F2.6) to identify the less frequent traces which are less likely to be considered if discovered manually. This will also benefit from being tested before implementation with scenario simulation (F5.4).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Monitoring                                                         | <b>C2.8 Monitoring shortcomings.</b> Assuming that impacted processes generate data, this BPCC can be satisfied by monitoring events in real-time (F4.11), checking compliance rules against the full population of data (F4.3), comparing “As-Is” and target models (F4.1) and checking conformance on a process graph (F4.5).<br><b>C2.9 Detecting process change.</b> This can be satisfied by comparing an “As-Is” model with a target model (F4.1) or by comparing variants of a process extracted from different time periods (F3.4).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <i>C3 Compliance checking – grouped by lifecycle phase</i>         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Design-time                                                        | <b>C3.1 Checking at design time.</b> Did not satisfy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



Table 7 continued

| BPCC category | BPCCs and the features that satisfy them                                                                                                                                                                                                                                                                                                                                    |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Run-time      | <b>C3.2 Checking at run time.</b> Features such as event-stream monitoring (F4.11), the conformance and compliance checking features (F4.1, F4.3, F4.7, F4.8) and compliance reporting features (F4.5-F4.6) support satisfying this BPCC, notwithstanding the caveat on the number of compliance rules to be checked and their potential impact on operational performance. |
| Audit         | <b>C3.3 Audit checking.</b> Applying process mining for <i>Audit checking</i> is well documented in the literature. The specific features that satisfy the BPCC – (F3.1, F3.2, F3.4, F4.1, F4.3, F4.5–F4.8) – have been discussed above.                                                                                                                                    |

Each BPCC is emphasised in bold and direct quotation is in bold

with process complexity and organizational issues than developing languages and techniques to formalize compliance rules and check them against a model at design time/run time. Regardless, there are both tactical and strategic issues for the research community to focus on. Tactical issues include: (i) deriving and rationalizing compliance rules from an event log; (ii) verifying control design and operating effectiveness; and (iii) visualizing controls and rules within a process model. From the business process perspective, process mining’s current capabilities should be extended to: (i) visualize compliance across multiple models; (ii) recommend options to simplify the business process environment; and (iii) quantify the magnitude and implications of both compliance requirements and process change. There is also a need to understand how the *C1 Compliance management* lifecycle and *C2 Business process management* lifecycle interact in practice and accommodate change in large, matrix-managed organizations.

At a more strategic level, the research opportunities include: (i) developing process mining ‘digital BPC twins’ to enable users from multiple functions to analyze the impact of their collective objectives; (ii) automating change management, to either update compliance rules or business processes; (iii) designing context-aware BPC solutions, that adapt to the users’ level of expertise; and (iv) developing preventative process mining techniques that prioritize recommendations based on reconciling conflicting objectives. Finally, from a business process perspective, banks have been trying to improve their processes for at least 20 years. The improvement teams have struggled to unravel the process ‘spaghetti’, and the banks’ response is to disband the team – “Every three to five years the banks get rid of all their process improvement people” (Adams et al. 2022). Why is this the case? Why can a bank not implement practices that have been proven to be successful in other industries?

*Implications for practitioners:* The implication for practitioners may appear trivial: fix the basics! In short, a process maturity uplift program is required (Rosemann and vom Brocke 2015). Successfully implementing these changes would facilitate the more strategic transformational change that the banks are undertaking. In many

industries, this recommendation would be a quick win but that is not the case in banking. Implementing this type of change on end-to-end processes that span multiple business units with no accountable owner, with complex legacy technology, with continuous operating model change, in an environment characterized by accelerating, transformational change, is exceptionally difficult.

*Implications for solution vendors:* For vendors, the primary focus should be on consolidating existing capabilities into a dedicated ‘low code/no code’, BPC solution which can: (i) ingest/link to control registers; (ii) automatically generate compliance rule templates; (iii) visualize controls in the discovered process; (iv) incorporate tools to support the user to identify redundant, duplicated and conflicting compliance rules, (v) extend the scope of predictive and preventative monitoring to cover all process perspectives, (vii) provide reporting across multiple processes, and (viii) enhance BPC simulation and ‘what-if’ analysis for design-time modeling.

*Implications for regulators:* Commissioner Hayne makes fourteen recommendations about the regulatory model and two recommendations about simplifying legislation (Hayne 2019). These recommendations are aligned to the findings in this study.

5.3 Threats to Validity

Our analysis of the BPC and process mining literature, contextualized within the Australian banking industry, has some limitations. There is a risk that the findings cannot be inferred for other banks and financial services participants, either within Australia or other jurisdictions. The context is derived from publicly available regulatory reports and documents, there may be a level of insight that is not accessible, e.g., the bank perspective, however, this was partially addressed in the study by Adams et al. (2022).

We assessed the solutions of five representative commercial vendors, their respective solutions are evolving rapidly and there may be functionality in the pipeline capable of addressing some of the gaps identified. There may also be other vendors whose solutions address some of these gaps.

**Table 8** Mapping features to BPCCs by BPCC category (2 of 2)

| BPCC category                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | BPCCs and the features that satisfy them                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>C4 Compliance analysis</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Reporting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p><b>C4.1 Measuring what matters.</b> As well as a range of pre-defined metrics covering frequency (F2.7) time (F2.8), and others, e.g., resources (F2.9), the ability to customize metrics/KPIs (F6.4) and set KPI thresholds (F6.5) will allow users to satisfy this BPCC. Capturing accurate metrics and KPIs is dependent on a range of <i>F1 Data management</i> features to incorporate attributes (F1.8), business hours and work patterns, e.g., shifts and holidays (F1.9), rules for specifying the event order where the timestamps are identical (F1.10) and ensuring both start and end timestamps are captured (F1.11)</p> <p><b>C4.2 Aligning reporting.</b> The ability to create custom dashboards (F6.3) that are aligned to the organizational hierarchy (F5.3), incorporate complex charting (F6.6), geographical mapping (F6.7), and be able to export reports (F6.1), charts and tables (F6.2) satisfies this BPCC</p>                                                                  |
| <i>C5 Compliance enactment (violation management)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Recover                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p><b>C5.1 Creating timely alerts.</b> This BPCC is satisfied by generating alerts (F4.10) that are linked to the organizational hierarchy (F5.3)</p> <p><b>C5.2 Handling multiple violations.</b> This is satisfied by listing all violations to record and document the violations accurately (F4.6)</p> <p><b>C5.3 Prioritizing the right violations.</b> Did not satisfy</p> <p><b>C5.4 Finding root cause.</b> This is primarily satisfied by the root cause analysis features (F3.3, F4.9) based on the list of violations generated by F4.6. The capability can be extended by including features such as process animation (F2.4) to help visualize a violation, or F4.5 to visualize conformance in a process graph, or variant analysis to compare “clean” and violated models (F3.4), as well as drilling down to individual case details (F3.6) or applying task mining to drill down to key stroke level (F5.6)</p> <p><b>C5.5 Recovering satisfactorily from violations.</b> Did not satisfy</p> |
| Prevent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p><b>C5.6 Predicting violations accurately.</b> This is satisfied by F5.1</p> <p><b>C5.7 Preventing repeat violations.</b> For real-time prevention, this is satisfied by prescriptive analytics (F5.2) and execution support (F5.7). To prevent repeat occurrences of violations discovered post execution, a combination of custom dashboards (F6.3), violation lists (F4.6), root cause analysis (F3.3, F4.9) and execution support (F5.7) will satisfy this BPCC</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <i>C6 Business support</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| User                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p><b>C6.1 Supporting the user.</b> Features that help satisfy this BPCC include: i) being able to visualize a process graph (F2.1) and in-graph conformance checking (F4.5), ii) the ability to view the process at different levels of abstraction, from a high-level end-to-end view to an individual, outlying case (F2.6), iii) the ability to save common rule patterns in compliance rule templates (F4.4), iv) multi-lingual support (F6.9), v) features that support sharing and collaboration (F6.10), vi) the availability of user training and support (F7.6), and vii) affiliated partnerships, preferably available in country for local support (F7.7)</p>                                                                                                                                                                                                                                                                                                                                      |
| Technical                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p><b>C6.2 Integrating securely.</b> This BPCC is satisfied by a combination of the full range of data management features (F1.1-F1.11) and the range of deployment models (F7.1), e.g., on-premises or cloud-based, user access management (F7.3), user authentication (F7.4), and the ability to generate audit logs (F7.5). Features such as data masking (F1.5) are particularly important given the sensitivity of the underlying data</p> <p><b>C6.3 Scaling across the enterprise.</b> This is satisfied by F7.2 and the fact that each of the solutions is already deployed in a commercial banking environment</p> <p><b>C6.4 Demonstrating real-world use.</b> The analyzed solutions are already deployed in a commercial banking environment</p>                                                                                                                                                                                                                                                   |
| <i>C7 Organizational</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| While the solution features cannot satisfy these BPCCs directly, they can assist management to address them. We provide examples below.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p><b>C7.1 Addressing business model complexity.</b> The extensive scope of the data management features (F1.1-F1.11) and the deployment model options (F7.1) indicate that business model complexity will not prevent deployment, and while it may not solve the complexity problem, it will help management visualize the fragmentation, which business units and ecosystem partners support a process, where they are located, which resources are required, and their reporting lines</p> <p><b>C7.2 Implementing enterprise-wide governance.</b> Aligning the <i>F5 View, monitoring and reporting</i> features (F6.1-F6.10) to the organizational hierarchy (F5.3) will provide management with a view of where BPC is actively managed, in a standardized way, integrating with other enterprise risk management systems</p> <p><b>C7.3 Changing an indifferent culture.</b> While improving monitoring overall should drive improved engagement, individual engagement can also be tracked by monitoring logins (F7.4) and the audit logs (F7.5)</p> <p><b>C7.4 Resolving conflicting objectives.</b> KPIs (F6.4), their thresholds/targets (F6.5) on custom dashboards (F6.3) aligned to the organizational hierarchy (F5.3) will provide an indication of conflicting objectives</p> <p><b>C7.5 Clarifying accountability.</b> Customized dashboards (F6.3) will also help clarify process ownership by following the organizational hierarchy (F5.3) from a resource associated with a rule violation to the executive that the resource ultimately reports to</p> <p><b>C7.6 Building staff knowledge.</b> The solutions will not address the knowledge gaps directly, but the business process management features (F2.1, F2.3, F2.4, F2.6-F2.9, F3.3, F3.7, F3.8) and the compliance enactment features, such as in-graph visualization of violations (F4.5) and conformance root cause (F4.9), as well as training (F7.6) and support from affiliated partners (F7.7) will help build process knowledge</p> <p><b>C7.7 Creating economic transparency.</b> Incorporating resource unit cost data into the models, along with the performance highlights feature (F2.3), process metrics (F2.7-F2.9), and the <i>F3 Process analysis and analytics</i> features (F3.1-F3.8) as well as conformance root cause analysis (F4.9) will not calculate the cost of compliance but will provide a level of economic transparency that is currently missing</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Each BPCC is emphasised in bold and direct quotation is in bold

**Table 9** Summary table showing the number of references, impact, and features by BPCC

| Challenge category                              | Total Refs n = 41 (RS)(IS) | Rating | # Features | Process mining feature satisfying the BPCCs         |
|-------------------------------------------------|----------------------------|--------|------------|-----------------------------------------------------|
| <i>C1 Compliance management</i>                 |                            |        |            |                                                     |
| C1.1 Eliciting convoluted requirements          | 21 (15) (6)                | DNS    | (9)        | (F1.1-F1.8, F5.5)                                   |
| C1.2 Interpreting complex requirements          | 21 (14) (7)                | DNS    | (1)        | (F4.4)                                              |
| C1.3 Handling different requirement types       | 12 (8) (4)                 | S      | 1          | F4.3                                                |
| C1.4 Checking all process perspectives          | 18 (18) (0)                | S      | 10         | F2.7-F2.9, F3.7, F3.8, F4.7, F4.8, F5.3, F5.6, F7.7 |
| C1.5 Formalizing rules accurately               | 18 (18) (0)                | S      | 2          | F4.3, F4.4                                          |
| C1.6 Implementing rules correctly               | 14 (13) (1)                | S      | 5          | F1.8, F4.4, F4.7, F4.8, F6.8                        |
| C1.7 Selecting the right checking approach      | 24 (18) (6)                | S      | 12         | F3.1, F3.4, F4.1-F4.8, F4.11, F6.8                  |
| C1.8 Rationalizing rules                        | 15 (13) (2)                | DNS    | (3)        | (F4.4, F6.3, F6.8)                                  |
| C1.9 Detecting requirements change              | 14 (9) (5)                 | DNS    | 0          | Did not satisfy                                     |
| <i>C2 Business process management</i>           |                            |        |            |                                                     |
| C2.1 Identifying fragmented processes           | 13 (7) (6)                 | S      | 11         | F1.1-F1.8, F1.11, F2.1, F2.6                        |
| C2.2 Extracting dispersed processes             | 11 (5) (6)                 | S      | 6          | F1.1-F1.4, F1.6, F1.7                               |
| C2.3 Discovering poorly documented processes    | 10 (4) (6)                 | S      | 4          | F2.1, F2.2, F2.6, F4.2                              |
| C2.4 Analyzing processes with sparse data       | 3 (2) (1)                  | S      | 6          | F2.1, F2.3, F2.6-F2.9                               |
| C2.5 Analyzing processes with poor quality data | 11 (4) (7)                 | S      | 7          | F1.1-F1.7                                           |
| C2.6 Working with diverse modeling languages    | 14 (13) (1)                | S      | 3          | F2.1, F2.2, F4.2                                    |
| C2.7 Implementing complex processes             | 16 (10) (6)                | S      | 3          | F2.1, F2.6, F5.4                                    |
| C2.8 Monitoring shortcomings                    | 13 (4) (9)                 | S      | 4          | F4.1, F4.3, F4.5, F4.11                             |
| C2.9 Detecting process change                   | 19 (12) (7)                | S      | 2          | F3.4, F4.1                                          |
| <i>C3 Compliance checking</i>                   |                            |        |            |                                                     |
| C3.1 Checking at design time                    | 15 (15) (0)                | DNS    | (4)        | (F3.1, F3.2, F4.3, F5.4)                            |
| C3.2 Checking at run time                       | 16 (16) (0)                | S      | 7          | F4.1, F4.3, F4.5-F4.8, F4.11                        |
| C3.3 Audit checking                             | 16 (12) (4)                | S      | 9          | F3.1, F3.2, F3.4, F4.1, F4.3, F4.5-F4.8             |
| <i>C4 Compliance analytics</i>                  |                            |        |            |                                                     |
| C4.1 Measuring what matters                     | 15 (8) (7)                 | S      | 9          | F1.8-F1.11, F2.7-F2.9, F6.4, F6.5                   |
| C4.2 Aligning reporting                         | 14 (8) (6)                 | S      | 6          | F5.3, F6.1-F6.3, F6.6, F6.7                         |
| <i>C5 Compliance enactment</i>                  |                            |        |            |                                                     |
| C5.1 Creating timely alerts                     | 18 (12) (6)                | S      | 2          | F4.10, F5.3                                         |
| C5.2 Handling multiple violations               | 15 (12) (3)                | S      | 1          | F4.6                                                |
| C5.3 Prioritizing the right violations          | 10 (7) (3)                 | DNS    | (4)        | (F2.8, F4.10, F6.4, F6.5)                           |
| C5.4 Finding root cause                         | 15 (11) (4)                | S      | 8          | F2.4, F3.3, F3.4, F3.6, F4.5, F4.6, F4.9, F5.6      |
| C5.5 Recovering satisfactorily from violations  | 14 (9) (5)                 | DNS    | (1)        | (F4.4)                                              |
| C5.6 Predicting violations accurately           | 7 (5) (2)                  | S      | 1          | F5.1                                                |
| C5.7 Preventing repeat violations               | 8 (5) (3)                  | S      | 6          | F3.3, F4.6, F4.9, F5.2, F5.7, F6.3                  |
| <i>C6 Business support</i>                      |                            |        |            |                                                     |
| C6.1 Supporting the user                        | 17 (15) (2)                | S      | 8          | F2.1, F2.6, F4.4, F4.5, F6.9, F6.10, F7.6, F7.7     |
| C6.2 Integrating securely                       | 10 (9) (1)                 | S      | 15         | F1.1-F1.11, F7.1, F7.3-F7.5                         |

**Table 9** continued

| Challenge category                           | Total Refs n = 41 (RS)(IS) | Rating | # Features | Process mining feature satisfying the BPCCs                           |
|----------------------------------------------|----------------------------|--------|------------|-----------------------------------------------------------------------|
| C6.3 Scaling across the enterprise           | 15 (6) (9)                 | S      | 1          | F7.2                                                                  |
| C6.4 Demonstrating real-world use            | 11 (11) (0)                | S      | All        | Solutions assessed have banking clients                               |
| <i>C7 Organizational</i>                     |                            |        |            |                                                                       |
| C7.1 Addressing business model complexity    | 11 (4) (7)                 | E      | 12         | F1.1-F1.11, F7.1                                                      |
| C7.2 Implementing enterprise-wide governance | 8 (2) (6)                  | E      | 11         | F5.3, F6.1-F6.10                                                      |
| C7.3 Changing an indifferent culture         | 9 (2) (7)                  | E      | 2          | F7.4, F7.5                                                            |
| C7.4 Resolving conflicting objectives        | 6 (1) (5)                  | E      | 4          | F5.3, F6.3-F6.5                                                       |
| C7.5 Clarifying accountability               | 8 (0) (8)                  | E      | 2          | F5.3, F6.3                                                            |
| C7.6 Building staff knowledge                | 10 (2) (8)                 | E      | 14         | F2.1, F2.3, F2.4, F2.6-F2.9, F3.3, F3.7, F3.8, F4.5, F4.9, F7.6, F7.7 |
| C7.7 Creating economic transparency          | 15 (6) (9)                 | E      | 13         | F2.3, F2.7-F2.9, F3.1-F3.8, F4.9                                      |

Key: RS = 32 research sources, IS = 9 industry sources, S = Satisfied, DNS = Did Not Satisfy, E = Enabled. Feature ratings, their

Key cont.: corresponding feature IDs and counts that appear in parentheses represent BPCCs satisfied after the paradigm shift workaround

Our analysis draws from studies presented in the BPC and process mining meta-reviews. While many of these studies addressed a specific problem, we extrapolated a generalization of the proposed solutions to determine how they could be applied to address BPCCs in the Australian banking context. For example, we assumed that an online conformance checking solution discussed in a maritime safety context (Maggi et al. 2011) can also be applied in a banking context.

## 6 Conclusion

While there are known challenges from the research community's perspective, the industry perspective, and particularly that of the industry's regulators, is largely absent. This study was enabled by the publication of four recent, detailed reports covering compliance-related events in the Australian banking industry. Supplementing this information with insights from practitioners and consultancy groups active in the industry, we developed a comprehensive list of 41 BPCCs across seven categories that captured both perspectives. The relative importance of each BPCC differs by perspective.

By investigating five commercial process mining solutions, we explored the potential for process mining to address the BPCCs. We identified 63 features across five categories and mapped these features to the BPCCs. We found that the features "Satisfied" 27 of the BPCCs and "Enabled" the *C7 Organizational* BPCCs. The commercial solutions had their limitations – the approach to satisfying the BPCCs was potentially cumbersome, and, of the seven

BPCCs that were rated "Did not satisfy", three were fundamental to a comprehensive BPC solution. However, by changing the paradigm, i) taking an organization's ROCR as the compliance requirements' source, and ii) using event logs for design-time and run-time compliance checking, we believe there is a pathway to address the BPCCs and improve compliance. While this paper leveraged Australian banking as its context, there is nothing to suggest that the findings cannot be generalized.

There are implications for all stakeholders in this study. For researchers, a closer alignment to industry's needs is required. For practitioners, the business process environment must be simplified. For the vendors, incorporating their existing capabilities into a BPC-oriented module and enhancing these with design-time model checking solutions will provide a roadmap to a comprehensive BPC solution.

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